

Tutor Handbook

Optimal Market Making for Cryptocurrency Options

The Guéant–Lehalle–Fernandez–Tapia Framework and its Extensions

A 10-Week Summer Research Program

North Carolina State University

Prepared for distribution to teaching assistants,
recitation leaders, and graders

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How to Use This Handbook

This handbook is the operational companion to the student-facing syllabus. It is intended for tutors — teaching assistants, recitation leaders, and graders — who will deliver the day-to-day instruction of the program under the high-level direction of the program mentor. It is not a substitute for the syllabus; you should read the syllabus first so that you understand the arc of the program from the student's perspective.

The handbook is organized into four parts. Part I is your orientation: it covers your role, your relationship with the mentor, the weekly rhythm of the program, and the policies you will need to enforce. Part II is the heart of the handbook: a week-by-week teaching guide that includes lecture emphasis, common student confusions, recitation lesson plans, problem set solution sketches, and grading rubrics. Part III provides milestone evaluation rubrics that you will use to grade the three software milestones. Part IV is an administrative reference covering academic integrity, late work, accommodations, and crisis escalation.

If you are new to the program, read Parts I and IV in full before the first week. Refer to the Part II week section the weekend before each week, and to the relevant Part III rubric in the week before each milestone deadline. Part II is intentionally written so that you can read one week in roughly thirty minutes of focused preparation; the goal is utility, not exhaustiveness.

Conventions Used in This Handbook

Throughout the document, the following conventions apply:

- Boldface marks the single most important point in a section.
- Italics indicate a quotation from the student-facing syllabus, included for cross-reference.
- A “red-flag” icon (annotated in the heading) marks situations where you should escalate to the mentor rather than handling the issue yourself.
- Grading rubrics are presented as tables with point allocations summing to the total for that problem.

All time allocations are recommendations rather than requirements. Adapt to the cohort in front of you. The recitation lesson plans in particular are starting points; an engaged cohort may move faster, a struggling one slower.

Part I. Tutor Orientation

1.1 Your Role

Tutors are responsible for the day-to-day delivery of the program. Concretely, your responsibilities are: leading the weekly ninety-minute recitation, holding three weekly office hours of one hour each, grading weekly problem sets within seven days of submission, providing first-pass evaluation of milestone deliverables (final evaluation rests with the mentor), and flagging at-risk students to the mentor within twenty-four hours of identifying a concern.

Tutors are not responsible for delivering the main lectures (those are delivered by the mentor or invited faculty) or for unilaterally changing the syllabus, problem sets, or grading rubrics. If you believe a change is necessary, raise it with the mentor; the mentor decides.

1.2 Relationship with the Mentor

The program is mentored by Dendi Suhubdy. The mentor sets program direction, delivers the main lectures, and makes final decisions on grading disputes and milestone evaluations. Tutors operate under the mentor's direction but are expected to exercise independent judgment within the scope defined above.

Standing communication channels with the mentor:

- A weekly thirty-minute tutor-mentor sync, held on the same day each week, covering the previous week's recitation outcomes, problem set performance, and any flagged students.
- A shared Slack workspace for asynchronous questions. The mentor commits to responding within twenty-four hours during business days. Do not wait on Slack for time-critical issues; for those, call.
- An emergency escalation path: a direct phone line for crisis situations (student in acute distress, suspected academic integrity violation, data security incident). Use sparingly.

1.3 Weekly Rhythm

Your typical week, assuming standard cohort size of six to eight students, will look like this:

Day	Activity	Estimated Time
Sunday evening	Read the relevant Part II section of this handbook; prepare recitation materials; review the previous week's problem set submissions if not yet complete.	2–3 hours
Monday	Attend lecture 1 (you do not deliver, but you should be present to flag confusions for the recitation).	1.5 hours
Tuesday	Office hour 1.	1 hour
Wednesday	Attend lecture 2.	1.5 hours

Thursday	Lead recitation.	1.5 hours
Friday	Office hour 2.	1 hour
Saturday	Office hour 3 (typically Saturday morning to support weekend problem-set work).	1 hour
Weekend	Grade the previous week's problem set; prepare for the next week.	4–6 hours

Total weekly commitment is approximately fifteen to twenty hours, peaking during milestone weeks when you also participate in code review.

1.4 Office Hours Guidance

Office hours are the most variable part of your week. Some weeks you will have full attendance for the entire hour; other weeks no one will come. Both are normal. The following principles apply:

- Resist solving problems for students. The instinct is strong; resist it. Ask the student to explain what they have tried; ask what they think the next step should be; provide the smallest hint that unblocks them.
- Distinguish between conceptual confusion and implementation confusion. The remedy differs. For conceptual confusion, return to the lecture material; for implementation confusion, ask to see the code and read it together.
- Watch for the student who comes to every office hour with the same kind of confusion. That is a signal worth flagging to the mentor — they may benefit from one-on-one mentor time.
- If two students come with the same question on the same day, it is probably a class-wide gap. Note it and address it at the start of the next recitation.
- Office hours are not consulting. Do not give code review on milestone deliverables during office hours. That happens in scheduled code review sessions during milestone weeks. The reason: if you debug a student's milestone code in office hours, your judgment when grading is no longer independent.

1.5 Grading Standards

Consistency matters more than calibration of an individual grade. If you have two tutors, you must coordinate: grade the same three problems together for the first problem set to calibrate, and spot-check each other thereafter. The grading rubrics in Part II are starting points, not constraints; if you find a rubric is misleading once you have seen real submissions, raise it with the mentor and the rubric will be updated.

Specific standards:

- Mathematical derivations are graded on correctness and clarity. Skipping a non-trivial step costs at most one point; getting the wrong answer with the right structure typically receives roughly half credit.
- Implementation problems are graded on correctness, reproducibility, and code quality, in that order. Code that runs and produces the right answer with poor style is graded above code that does not run at all.

- Discussion problems are graded on substantive engagement. A discussion that takes a defensible position and supports it with the material from lecture receives full credit; a discussion that restates the question without engaging receives zero.

Return graded work within seven days of the submission deadline. Provide written feedback on at least one problem per problem set, even when full marks are awarded.

1.6 Tutor Calibration Session

Before the program begins, the mentor holds a half-day tutor calibration session. The agenda is: review the syllabus; walk through one full week of teaching notes from this handbook together; grade a sample problem set (provided by the mentor) as a group; review the milestone rubrics; and resolve any open questions. Attendance is mandatory.

Part II. Week-by-Week Teaching Notes

Each of the ten weekly sections below is structured identically: the spine of the week, lecture emphasis notes, common student confusions with suggested handling, a recitation lesson plan, and the problem set solution sketches with grading rubric. Read the section for the upcoming week in full before the Sunday preparation block.

Week 1. Foundations of Market Microstructure

THE SPINE OF THIS WEEK

Students must leave this week able to think of a market maker as a continuously quoting participant whose existence is justified by the spread, and able to read a limit order book without having to look up vocabulary. Everything in the next nine weeks builds on this.

Lecture Emphasis

Below are notes for each of the two lectures. You do not deliver these; you attend them. Your job is to track where the cohort got lost so you can address it in recitation.

Lecture 1 — Limit Order Book Mechanics

The lecture is intentionally not mathematical; its goal is vocabulary and institutional fluency. The mentor will move quickly through order types and matching rules. The important conceptual point, which students often miss because it is delivered casually, is that price-time priority creates a queue at each price level and that being earlier in that queue is a real economic asset. This will matter in week 5 when fill probabilities are calibrated.

Watch for student reactions at:

- When the mentor introduces post-only orders. Students often confuse these with limit orders and do not understand the economic motivation.
- When the mentor explains the difference between L2 and L3 data. Students with prior equity-finance exposure often think L2 is the only thing that exists.

Lecture 2 — Spread Decomposition and the Role of the Market Maker

Three models are covered in one lecture: Roll, Glosten–Milgrom, and Kyle. This is fast. The mentor will likely pause to draw the Glosten–Milgrom decision tree on the board, and that drawing is the most important artifact of the lecture. Encourage students to copy it exactly. The Kyle model is included for completeness and is the one students consistently struggle with; the mentor may downplay it depending on time.

Watch for student reactions at:

- The transition from Roll to Glosten–Milgrom. Students sometimes think Glosten–Milgrom is just a fancier version of Roll. It is not: they answer different questions.
- The Kyle equilibrium. Several students will be lost. Do not worry; week 1 problem 5 does not require Kyle.

Common Student Confusions

Confusions you should expect to see in office hours and recitation. The right-hand column gives suggested handling.

Confusion	How to Handle
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Why does the market maker quote both sides? Why not just buy low and sell high reactively?	Walk through the time horizon. A market maker commits in advance; a reactive trader commits in response. The bid-ask spread is the compensation for committing in advance under uncertainty.
Aren't all orders limit orders, since a market order is a limit order at infinity?	Mathematically yes; institutionally no. Market orders are explicitly different objects on real exchanges because they have different fee schedules and matching priority. Refer to the Coincall fee schedule for a concrete example.
L2 and L3 — when does it matter which you use?	L3 matters when you care about queue position (your specific order, not aggregate depth). For the calibration work in week 5, L2 is sufficient because we model aggregate intensity rather than individual queue dynamics.
The Roll estimator gives a different number than the observed spread. Bug in my code?	Almost certainly not. The Roll estimator estimates the unobserved component of the spread, which differs from the quoted spread by adverse-selection and price-impact effects. The discrepancy is the interesting finding, not a bug.

Recitation Lesson Plan

Goal: Every student leaves recitation with a working Python class that reconstructs an L2 order book from a stream of events. This is the foundation for everything in week 5 and beyond.

Materials needed:

- A small (one-hour) Coincall BTC-PERP event stream in JSON, distributed to students in advance.
- A reference solution to the order book reconstruction class, available to the tutor only.
- A laptop projector or shared screen.

Suggested timing for the ninety-minute session:

Minutes	Activity
0–10	Greet the cohort; recap the goal of the week; quick poll on which parts of the two lectures landed well and which did not.
10–20	Live demo of loading the data file in Python; show the shape of an order event; demonstrate one minute of order book reconstruction manually on the whiteboard.
20–60	Pair programming. Students work in pairs to extend the partial scaffold (provided) into a working reconstructor. Circulate continuously. The most common bug is forgetting to clean up empty price levels after a cancel; if you see this in one pair, announce it to the room.
60–75	Reconvene. One pair presents their reconstructor to the room; the cohort debugs any

	final issues collectively.
75–90	Wrap-up. Address questions from the lectures that were flagged during the cohort poll. If time remains, sketch how this code will be extended in week 5 to support a fill simulator.

Recitation-leader notes:

- If most pairs are not converging by minute 50, stop the pair programming and walk through the reference solution together. It is better that everyone leaves with working code than that some struggle through a half-finished version.
- Save the working code; students will reuse it throughout the program.

Problem Set Solutions and Rubric

Solution sketches with point allocations. Tutors are expected to develop their own complete solutions before grading; the sketches below are calibration aids, not solutions manuals.

#	Pts	Key result expected	Common errors and partial credit
1	10	Working Python class that reconstructs the L2 book event by event; a plot of mid-price and quoted spread over one hour of data.	Most common error: not handling cancels on empty price levels. -2. Second most common: incorrect handling of the trade event (decrementing the wrong side). -3. If code does not run at all: at most 3 points for an apparent design.
2	8	Roll estimator computed; comparison plot with the quoted spread; brief discussion of the discrepancy that correctly identifies the role of asymmetric information.	Computing the Roll estimator with the wrong sign convention (it should be the square root of negative serial covariance, and students often miss the sign). -3. Discussion that simply notes the discrepancy without engaging: -3.
3	10	Analytical derivation of the Glosten–Milgrom quote-setter's bid and ask. Should arrive at expressions involving the probability of informed trading and the prior on asset value.	Skipping the Bayesian update step: -3. Confusing the bid with the ask: -2. Failing to acknowledge that the spread is endogenously determined by the indifference condition: -2.
4	8	Simulation of Glosten–Milgrom with chosen parameters; a plot of the public belief converging toward true asset value over a sequence of trades.	Plotting only one realization rather than averaging: -2. Using a non-rational prior (e.g., starting the simulation with the correct belief): -2.
5	9	Realized spread and price impact computed	Computing realized spread with the

		over a five-minute window; discussion of order-flow toxicity that engages with the empirical numbers obtained.	wrong sign of trade direction: -3. Discussion that does not reference the actual numbers from the data: -3.
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Week 2. The Avellaneda–Stoikov Model

THE SPINE OF THIS WEEK

Students must leave this week with a complete worked example of the stochastic-control pipeline: dynamics, HJB, ansatz, ODE reduction, closed-form solution. Everything in weeks 3–5 will be variations on this template.

Lecture Emphasis

Below are notes for each of the two lectures. You do not deliver these; you attend them. Your job is to track where the cohort got lost so you can address it in recitation.

Lecture 1 — Setup and Dynamics

The lecture establishes the modeling choices. The two most important: arithmetic Brownian motion for the mid-price (not geometric — this matters; the linearity simplifies the HJB enormously), and exponential utility (which decouples the value function from wealth). Make sure students appreciate that these are choices made for tractability, not laws of nature.

Watch for student reactions at:

- When arithmetic versus geometric Brownian motion comes up. Some students will object; they should be told this is a local approximation valid over the trading horizon.
- The introduction of the intensity function. Students often want to immediately ask where it comes from empirically; defer this to week 5.

Lecture 2 — HJB Derivation and Solution

The technical centerpiece of the week. The separation ansatz $V(t, x, S, q) = -\exp(-\gamma(x + qS)) \theta(t, q)$ is the move students must internalize, because it appears in every subsequent control problem in the program. The mentor will derive the reduced ODE system carefully; this is the place where attendance matters most.

Watch for student reactions at:

- The moment the ansatz is introduced. If students do not understand why this is the right ansatz, they will not understand any subsequent HJB reduction. Plan to revisit in recitation.
- The boundary terms. Students often miss that the terminal condition feeds into the asymptotic expansion.

Common Student Confusions

Confusions you should expect to see in office hours and recitation. The right-hand column gives suggested handling.

Confusion	How to Handle
Why exponential utility? It seems unrealistic to assume traders have CARA preferences.	It is unrealistic; this is purely a tractability choice. CARA decouples the value function from wealth, which makes the problem solvable. Without it, you cannot make

	analytical progress. The mentor's view: model market makers' "effective" risk preferences over a short horizon, not their lifetime preferences.
The ansatz seems to come out of nowhere.	It does, in the sense that there is no algorithm for guessing ansätze. But it is principled: under exponential utility plus arithmetic mid-price dynamics, the value function inherits a multiplicative structure across the state variables, which the ansatz reflects. Students should accept this on faith for now; we will see the same pattern in three more weeks.
Why is the optimal half-spread symmetric in inventory, but the reservation price asymmetric?	Different quantities. The reservation price is the indifference point (asymmetric because being long shifts your indifference point lower). The optimal half-spread is the width of the quoting band around that reservation point (approximately symmetric to first order). They are independent design choices.
My P&L is huge in simulation. Is that realistic?	No. The Avellaneda–Stoikov simulation assumes you are alone in the market and the order flow is exogenous. Real markets have competitors and the order flow responds to your quotes. Weeks 5 and 9 calibrate to real Coincall data and the P&L estimates become more realistic.

Recitation Lesson Plan

Goal: Every student completes the HJB derivation on the whiteboard with the cohort, then implements the simulation and runs sensitivity analysis on the risk-aversion parameter.

Materials needed:

- Whiteboard or large shared canvas for the derivation.
- A Python notebook scaffold for the simulation, with the dynamics and intensity functions implemented but the quote-computation left blank.

Suggested timing for the ninety-minute session:

Minutes	Activity
0–30	Group derivation on the whiteboard. The tutor leads, but students contribute each step. Aim to complete the derivation from dynamics through reduced ODE in thirty minutes.
30–60	Pair programming: students fill in the quote-computation function in the provided notebook. Most pairs will get this working in twenty minutes; remaining time is for the simulation.
60–80	Run the sensitivity analysis: each pair varies γ over three orders of magnitude and

	reports back the inventory variance and Sharpe ratio. Aggregate results on the projector.
80–90	Discussion. What does the sensitivity plot tell us about the model? Where would you set γ for a real market making operation? Tie this to problem 4.

Recitation-leader notes:

- The whiteboard derivation is the single most important hour of the program for many students. Take your time. If you do not finish in thirty minutes, continue into the implementation slot — the implementation can be deferred to the problem set if necessary.

Problem Set Solutions and Rubric

Solution sketches with point allocations. Tutors are expected to develop their own complete solutions before grading; the sketches below are calibration aids, not solutions manuals.

#	Pts	Key result expected	Common errors and partial credit
1	15	Complete HJB derivation, separation ansatz, reduction to ODE system. Every step justified.	Missing the $It\hat{o}$ term for the underlying: -4. Incorrect treatment of the jump in the wealth process: -4. Wrong sign in the cash dynamics: -2.
2	10	Closed-form asymptotic solution: reservation price and half-spread. Should arrive at $r = S - q\gamma\sigma^2(T-t)$ and $\delta = (1/\gamma)\log(1+\gamma/\kappa) + (\gamma\sigma^2/2)(T-t)$.	Wrong limit assumption: -2. Algebra errors in the half-spread: -3.
3	12	Working AS engine in Python; one-day simulation with stated parameters; P&L distribution comparing optimal vs symmetric.	Symmetric baseline not actually symmetric (drift in quotes): -3. Comparing wrong distributions: -3. Code does not run: at most 4 points for design.
4	10	Sensitivity analysis on γ ; plots of inventory variance and Sharpe vs γ ; identification of optimal regime.	Only one γ value tested: -5. Wrong axis scaling (linear when log is required): -2.
5	8	Discussion of three empirical violations of the independence assumption. Should engage with phenomena like price impact, adverse selection, and competitor effects.	Listing without explanation: -3. Failing to discuss consequences for the strategy: -3.

Week 3. Stochastic Optimal Control in Continuous Time

THE SPINE OF THIS WEEK

Students should leave with the abstract template — dynamic programming principle, HJB equation, verification theorem — and the ability to recognize when a problem fits the template. The Merton problem is the second worked example; it should reinforce, not introduce, the pattern from week 2.

Lecture Emphasis

Below are notes for each of the two lectures. You do not deliver these; you attend them. Your job is to track where the cohort got lost so you can address it in recitation.

Lecture 1 — Dynamic Programming and the HJB Equation

The abstract version of last week. The mentor will derive the HJB equation in generality, treating the previous week's Avellaneda–Stoikov as a special case. Students who struggled last week often have an epiphany here; students who breezed through last week sometimes find this lecture unexciting. Both reactions are normal.

Watch for student reactions at:

- The infinitesimal generator. Students with weak stochastic-calculus background often need this re-explained.
- Viscosity solutions. The mentor will mention them briefly. Do not let students get stuck on this; viscosity solutions are not central to the program.

Lecture 2 — Verification Theorems and Tractable Utility Functions

The verification theorem is the missing piece from week 2: it tells you that the solution you find by ansatz is actually the value function. The lecture will state the theorem precisely and sketch the proof. The CARA-versus-CRRA discussion that closes the lecture is also important; some students will only fully understand the choice of CARA in week 2 after seeing the comparison with CRRA in Merton's problem.

Watch for student reactions at:

- Smoothness conditions in the verification theorem. Students sometimes ask whether these conditions hold for the GLFT problem — they do, but it takes work to show; defer.

Common Student Confusions

Confusions you should expect to see in office hours and recitation. The right-hand column gives suggested handling.

Confusion	How to Handle
Why is the HJB equation a necessary condition? Couldn't there be other optimal policies?	The HJB encodes the dynamic programming principle, which is necessary by definition (any optimal policy must be optimal at every later time conditional on the current state). The verification theorem provides sufficiency.

The Merton problem under CRRA has wealth in the policy, but Avellaneda–Stoikov under CARA does not. Why?	This is exactly the point. CARA exponential utility makes the marginal utility of wealth constant, which makes the optimal policy independent of wealth. CRRA does not have this property; the optimal portfolio share is constant in wealth but the dollar amount scales. Encourage the student to write out both derivations side by side.
What is the difference between a verification theorem and an existence theorem?	Verification: "if you have a smooth solution, it is the value function." Existence: "a solution exists." In our program we always have a candidate solution (we constructed it), so we only need verification.
Why do we ignore viscosity solutions when the textbook makes such a big deal of them?	Because all the problems in this program have smooth value functions. Viscosity solutions matter for problems with free boundaries or non-smooth payoffs (e.g., American options). Our problems are smooth enough that classical theory suffices.

Recitation Lesson Plan

Goal: Students complete the Merton problem under CARA from scratch, then under CRRA, and observe the structural difference between the two solutions.

Materials needed:

- Whiteboard for the derivations.
- A short reference sheet with the Itô formula, the dynamic programming principle, and the HJB equation, for students to consult.

Suggested timing for the ninety-minute session:

Minutes	Activity
0–10	Quick recap of HJB. Solicit one-sentence statements of the dynamic programming principle from three students.
10–45	Group derivation of the Merton problem under CARA. The tutor guides; students contribute each step.
45–75	Group derivation under CRRA. Move faster; students should be more confident by now.
75–90	Side-by-side comparison of the two solutions. Explicit discussion: why does wealth appear in one and not the other?

Recitation-leader notes:

- If the cohort is strong, you can save thirty minutes by doing CARA quickly and spending most of recitation on Almgren–Chriss (problem 3) instead. Adjust based on energy level.

Problem Set Solutions and Rubric

Solution sketches with point allocations. Tutors are expected to develop their own complete solutions before grading; the sketches below are calibration aids, not solutions manuals.

#	Pts	Key result expected	Common errors and partial credit
1	12	Merton problem under CRRA, with full derivation. Optimal consumption and portfolio share constant in wealth.	Incorrect ansatz: -4. Missing the HJB transversality: -3. Wealth appearing in the wrong place in the policy: -2.
2	10	Merton problem under CARA, with full derivation. Optimal portfolio dollar amount constant in wealth.	Same as problem 1, plus failure to articulate the structural difference: -2.
3	12	Almgren–Chriss optimal execution. Optimal trajectory derived; comparison with discrete-time result.	Skipping the calculus of variations step: -4. Wrong boundary conditions: -3.
4	10	Verification theorem in own words; identification of each assumption's role; counterexample with one assumption failing.	Restating the theorem verbatim: -5. Counterexample that does not actually fail the theorem: -3.
5	6	Two-paragraph essay on viscosity solutions and the obstacle problem.	Plagiarism from the textbook: 0. Missing the obstacle problem entirely: -3.

Week 4. The GLFT Model, Part I — Setup and HJB Derivation

THE SPINE OF THIS WEEK

GLFT is presented as Avellaneda–Stoikov with two generalizations: arbitrary intensity functions and bounded inventory. Students should be able to derive the GLFT HJB by analogy with week 2, identifying exactly where the generalizations enter.

Lecture Emphasis

Below are notes for each of the two lectures. You do not deliver these; you attend them. Your job is to track where the cohort got lost so you can address it in recitation.

Lecture 1 — The GLFT Setup

Conceptually a setup lecture rather than a technical one. The two key generalizations are clearly flagged, and the mentor will discuss why both matter in practice: arbitrary intensities because exponential intensities are an empirical fiction, and bounded inventory because no real market maker would tolerate unbounded position growth. Students should leave the lecture confident that the GLFT problem is well-posed.

Watch for student reactions at:

- When the inventory bounds $q_{\min}$, $q_{\max}$ are introduced. Students sometimes confuse these with the asymptotic limits.
- When the mentor defines admissible quote ranges (no quoting beyond the inventory bound). The treatment is subtle.

Lecture 2 — HJB Derivation in the GLFT Framework

The HJB derivation is closely parallel to week 2, with two new features: the intensity functions are general rather than exponential, and the inventory boundary requires careful treatment. The mentor will derive carefully but quickly, relying on the cohort's familiarity with the template from week 2.

Watch for student reactions at:

- The boundary conditions at $q_{\min}$ and $q_{\max}$. These are the most subtle part of the derivation; expect questions in the next office hour.

Common Student Confusions

Confusions you should expect to see in office hours and recitation. The right-hand column gives suggested handling.

Confusion	How to Handle
Why not just use exponential intensities and skip this generalization?	Exponential intensities are a strong assumption: they imply that the probability of fill drops geometrically with distance from mid. Empirically (we will see in week 5) the decay is often slower, especially in the tails. The

	generalization allows the model to fit reality.
What does 'arbitrary intensity function' actually mean? Are there restrictions?	Yes: the intensity must be positive, decreasing in quote distance, and satisfy some integrability conditions for the HJB to be well-posed. Read Section 2 of the GLFT paper for the precise statement. In practice, almost any reasonable parametric family works.
The boundary conditions at $q_{\min}$ and $q_{\max}$ are confusing. Are they the same as the terminal condition?	No. Terminal condition: what happens at time T . Boundary condition: what happens at the inventory limits. They are independent. The boundary conditions encode that you stop quoting on the side that would push you further into the constraint.
How is the asymptotic regime defined? Just $T \rightarrow \infty$?	It is: the asymptotic regime is where $T - t$ is large enough that the terminal effects are negligible. We will use a closed-form expression for the asymptotic quotes in week 5; the technicalities of the limit are sketched in the paper but not central.

Recitation Lesson Plan

Goal: The cohort re-derives the GLFT HJB step by step on the whiteboard, with explicit attention to the points where the generalization from Avellaneda–Stoikov enters.

Materials needed:

- Whiteboard or shared canvas.
- A copy of the week 2 whiteboard derivation, projected on a separate screen for side-by-side comparison.

Suggested timing for the ninety-minute session:

Minutes	Activity
0–15	Recap of last week. Solicit a one-sentence statement of the difference between AS and GLFT from each student.
15–60	Group derivation of the GLFT HJB. At each step, ask explicitly: "is this step the same as Avellaneda–Stoikov, or different?" When different, why?
60–80	Discussion of intensity function choices. The tutor presents three families (exponential, power-law, logistic) and asks the cohort which they would calibrate to which kind of order flow.
80–90	Wrap-up. Preview of week 5: closed-form solution in the asymptotic regime.

Recitation-leader notes:

- The side-by-side comparison with week 2 is the pedagogical device that makes this recitation work. Students who can articulate the difference between AS and GLFT have understood the material; those who cannot have not.

Problem Set Solutions and Rubric

Solution sketches with point allocations. Tutors are expected to develop their own complete solutions before grading; the sketches below are calibration aids, not solutions manuals.

#	Pts	Key result expected	Common errors and partial credit
1	15	Complete GLFT HJB derivation with boundary conditions. Identification of the precise step where Λ regularity is needed.	Missing boundary conditions: -5. Treating Λ as exponential without explicitly noting the generalization: -3.
2	8	Algebraic reduction of GLFT to AS HJB under $\Lambda(\delta) = A \exp(-\kappa\delta)$.	Algebra errors: -2 each. Failure to relax the inventory constraint: -3.
3	10	Three intensity families plotted and compared, with discussion of the behavioral assumptions each encodes.	Plotting only without discussion: -4. Discussion that does not connect to trader behavior: -3.
4	8	Inventory dynamics derived under generic Λ ; expected change computed.	Treating the jump term incorrectly: -3. Computing variance instead of expectation: -2.
5	9	One-page summary of modeling differences between AS and GLFT, distinguishing mathematical from notational differences.	Failing to distinguish substantive from notational: -4. Going over one page: -1 (we are training for industry research notes).

Week 5. The GLFT Model, Part II — Asymptotic Solution, Implementation, and Variants

THE SPINE OF THIS WEEK

Two distinct deliverables this week: students must understand the closed-form asymptotic solution and submit Milestone 1, the linear-asset GLFT engine. The variants survey is self-study; do not let it eat lecture or recitation time.

Lecture Emphasis

Below are notes for each of the two lectures. You do not deliver these; you attend them. Your job is to track where the cohort got lost so you can address it in recitation.

Lecture 1 — The Asymptotic Regime and Closed-Form Quotes

The technical centerpiece of the program so far. The change of variable $u(q) = \theta(t, q)/\theta(t, q+1)$ linearizes the coupled ODE system into a linear eigenvalue problem. Students will be told this is "the trick that makes the paper work" — and it is. The mentor will derive it carefully and may extend the lecture if students are following well.

Watch for student reactions at:

- The change of variable. If students do not understand why this particular ratio works, they will not understand the closed-form solution. Re-derive in office hours if needed.
- The Perron–Frobenius theorem. Some students will not have seen it before; the lecture will not stop to explain. Plan a five-minute aside in recitation.

Lecture 2 — Numerical Implementation and Validation

The transition from theory to code. The mentor will demonstrate the implementation live: setting up the linear system, calibrating intensities from data, and running a simulation. This is also the lecture where the milestone deliverable is formally announced and the rubric is reviewed.

Watch for student reactions at:

- The intensity calibration step. Students often want to use a simple curve fit but will get poor results; maximum likelihood is required and the mentor will show why.

Common Student Confusions

Confusions you should expect to see in office hours and recitation. The right-hand column gives suggested handling.

Confusion	How to Handle
Why is the change of variable $u(q) = \theta(t, q)/\theta(t, q+1)$ and not the natural log?	Because the ODE system has a multiplicative coupling between $\theta(t, q)$ and $\theta(t, q+1)$ that becomes linear under the ratio. Taking logs first does not produce a linear system; taking the ratio first does. This is one of those

	choices that is hard to motivate in advance and obvious in retrospect.
My linear system is singular. What did I do wrong?	Most likely you have not enforced the boundary condition properly at $q_{\max}$. The asymptotic linear system needs an explicit normalization (e.g., $u(q_{\max}) = 1$) to remove the multiplicative gauge.
The calibrated intensity functions look implausible — should I be worried?	Probably not at first. Crypto fill data is noisy and the maximum likelihood fit on a small sample can give surprising parameters. Try fitting on one week, then two weeks, then four. The estimates should stabilize.
My backtest P&L is negative. Is the model wrong?	Possibly, but more likely: (a) the spread is too tight relative to the calibrated intensities, (b) the inventory penalty is too small, or (c) the data window contained a regime shift. Have the student check all three before suspecting the model.

Recitation Lesson Plan

Goal: Students complete the change-of-variable derivation, code review one student's milestone-in-progress, and identify any remaining bugs before the milestone deadline.

Materials needed:

- Whiteboard.
- A projector for live code review (one student's code, with their permission).
- A list of expected milestone bugs, prepared by the tutor in advance based on office hours that week.

Suggested timing for the ninety-minute session:

Minutes	Activity
0–25	Whiteboard derivation of the change of variable, leading to the linear eigenvalue problem. Aside on Perron–Frobenius (five minutes within this block).
25–70	Live code review. One student volunteer; their code projected. The cohort identifies issues. The tutor moderates; the volunteer should leave with a working implementation.
70–85	Open Q&A on the milestone. Common questions: what counts as "non-trivial fit" for the calibration? What if my latency is $150\mu\text{s}$ instead of $100\mu\text{s}$? How long should the technical report be?
85–90	Wrap-up. Reminder of milestone deadline.

Recitation-leader notes:

- The code review is the high-value activity. It will reveal common bugs across the cohort, which you can address in the wrap-up.
- Do not let the variants survey discussion happen in recitation. It is self-study. If students want to discuss it, send them to office hours.

Problem Set Solutions and Rubric

Solution sketches with point allocations. Tutors are expected to develop their own complete solutions before grading; the sketches below are calibration aids, not solutions manuals.

#	Pts	Key result expected	Common errors and partial credit
1	12	Change-of-variable derivation; linear system in the asymptotic regime.	Wrong limit assumption: -3. Algebra errors in the change of variable: -2 each. Missing the normalization that fixes the gauge: -2.
2	8	Linear system solved numerically for stated parameters; half-spread plotted as a function of inventory.	Singular system (no normalization): -3. Wrong sign convention on quote distance: -2.
3	10	Reproduction of Figure 2 of GLFT (2013), with commentary on discrepancies.	No commentary: -4. Reproducing the wrong figure: -3 (read the assignment).
4	10	Maximum-likelihood calibration of intensities on one week of Coincall data; fitted parameters and goodness-of-fit reported.	Using OLS instead of MLE: -4. Failing to report goodness-of-fit: -3.
5	12	Full one-month backtest; all five performance metrics reported; comparison against symmetric baseline.	Missing the symmetric baseline: -4. Missing one or more metrics: -2 each. Reporting metrics without time-series plots: -2.
6	5	Two-page summary of Guéant (2016) Chapters 10–11; selection of one variant most relevant to crypto options.	Going over two pages: -1. Failing to select a specific variant: -2. Summary that is purely descriptive without engaging with relevance: -3.

Milestone Week Note

MILESTONE WEEK. Milestone 1 is due at the end of this week. See Part III, Section 1 for the full evaluation rubric. As a tutor, your role this week is heavier than usual: in addition to the recitation, you participate in milestone code review at the end of the week. Block out four hours of code review time in addition to your usual office hours.

Week 6. Options Pricing, the Volatility Surface, and Calibration

THE SPINE OF THIS WEEK

A two-track week. Conceptually: implied volatility surfaces and the constraints they must satisfy. Practically: calibrate SVI and SABR to Coincall data and discover how messy real crypto options data is. Students should leave knowing that calibration is half craft and half theory.

Lecture Emphasis

Below are notes for each of the two lectures. You do not deliver these; you attend them. Your job is to track where the cohort got lost so you can address it in recitation.

Lecture 1 — Greeks and the Volatility Surface

A brisk review of Greeks for students who have taken derivative pricing, plus a careful treatment of the no-arbitrage constraints on the volatility surface. The lecture will spend the most time on calendar and butterfly arbitrage, because these are the constraints that get violated by naive parameterizations. Vanna and volga are introduced; they will not be tested for full understanding until week 8.

Watch for student reactions at:

- The transition from the Black–Scholes formula to the Greeks. Students with weak Black–Scholes background will need this revisited in office hours.
- The arbitrage constraints. These are presented as inequalities; students sometimes miss that they are necessary and not sufficient.

Lecture 2 — Parameterization and Calibration

The lecture presents SVI and SABR as the two dominant parametric families and walks through calibration practically. The mentor will live-demo a calibration on a Coincall snapshot and show that the first attempt usually has stability issues; this is intentional, to teach the lesson that crypto calibration is hard.

Watch for student reactions at:

- The live calibration. Students should see it fail in interesting ways and watch the mentor diagnose. The diagnostic process is more valuable than the eventual fit.

Common Student Confusions

Confusions you should expect to see in office hours and recitation. The right-hand column gives suggested handling.

Confusion	How to Handle
Why two parameterizations? Why not just one?	They encode different priors. SVI is parameterized in implied variance and is natural for arbitrage-free conditions. SABR is parameterized in stochastic volatility model parameters and is natural when you want to extrapolate beyond the observed strikes. We will use both

	because they fail in different regimes.
My SVI calibration has huge residuals on far OTM puts. Bug?	Probably not. Far OTM puts in crypto have wide spreads and low volume; the mid-quote may be unreliable. Weighting the calibration by volume or by inverse spread is the standard remedy.
The SVI parameters look very different from one day to the next. Is this normal?	Yes and no. Day-to-day variation in market conditions causes real variation, but a well-calibrated parameter set should not jump wildly. If your fits are unstable, your initial guess is probably the issue; try seeding from yesterday's fit.
The Hagan SABR formula gives a different price than my Monte Carlo. Which is right?	Hagan is a small-time approximation. For short maturities (under three months) it is excellent; for longer maturities the error grows. Monte Carlo is the truth. The discrepancy is informative, not a bug.

Recitation Lesson Plan

Goal: Every student produces a working SVI and SABR calibration on a Coincall BTC options snapshot, with diagnostic plots, by the end of the session.

Materials needed:

- A specific Coincall BTC options chain snapshot (chosen by the tutor in advance, ideally one with reasonably clean data so that the calibration succeeds in recitation time).
- A starter notebook with the SVI and SABR pricing functions implemented, leaving calibration to the student.
- Reference diagnostic plots from the tutor's own calibration of the same snapshot.

Suggested timing for the ninety-minute session:

Minutes	Activity
0–10	Recap; show the chosen snapshot; explain what "successful" calibration looks like.
10–55	Pair programming. Each pair calibrates SVI first, then SABR. Tutor circulates and helps with optimizer convergence issues.
55–80	Each pair presents their fit with diagnostic plots. The cohort discusses sources of disagreement between pairs.
80–90	Discussion of next week's milestone and how the calibrated surface feeds into it.

Recitation-leader notes:

- If the chosen snapshot turns out to be too clean, the lesson about "crypto calibration is hard" will not land. Have a backup snapshot with one or two problematic strikes ready, and switch if recitation is going too smoothly.

Problem Set Solutions and Rubric

Solution sketches with point allocations. Tutors are expected to develop their own complete solutions before grading; the sketches below are calibration aids, not solutions manuals.

#	Pts	Key result expected	Common errors and partial credit
1	10	Vanna and volga derivations under Black-Scholes; finite-difference verification.	Algebra errors: -2 each. Failing to verify numerically: -3.
2	10	SVI implementation with arbitrage-free constraints; explicit demonstration of which constraints fail on a deliberately broken parameterization.	Implementing raw SVI without the constraint checks: -4. Not actually demonstrating violations: -3.
3	10	SVI calibration to one snapshot; fitted parameters, residuals, RMS error reported.	Reporting only the RMS error without per-strike residuals: -3. Optimizer failing to converge and student not noticing: -3.
4	10	SABR calibration to the same snapshot; comparison with SVI.	Comparison that does not engage with strengths and weaknesses: -3. Calibration that does not converge: -3.
5	10	One-month daily SVI calibration; time series of parameters; proposed smoothing scheme.	Producing the time series without commenting on stability: -3. Smoothing scheme that does not actually address the observed instability: -3.

Week 7. Fast Computation of Prices, Greeks, and Surfaces

THE SPINE OF THIS WEEK

The week is about latency. Students must internalize that a working but slow pricer is not enough for an options market maker. The two technical tools — characteristic function pricing and algorithmic differentiation — are the means; the end is a five-millisecond surface revaluation.

Lecture Emphasis

Below are notes for each of the two lectures. You do not deliver these; you attend them. Your job is to track where the cohort got lost so you can address it in recitation.

Lecture 1 — Characteristic Function Pricing

Carr–Madan and Fang–Oosterlee are both covered, but Fang–Oosterlee (COS) is the workhorse. The lecture spends more time on COS because it is what students will use for the milestone. Carr–Madan is included for historical context and because it remains common in legacy systems.

Watch for student reactions at:

- When the characteristic function of the Heston model is introduced. The expression is complex; students will not derive it themselves but should recognize its structure.
- The truncation interval in COS. Students sometimes choose it too narrow and get artifacts; the mentor will show this.

Lecture 2 — Algorithmic Differentiation and Vectorization

The conceptual lecture of the program for systems-minded students. The mentor will distinguish forward-mode from reverse-mode AD, demonstrate JAX live, and show the latency impact. C++ AAD via Enzyme is mentioned as advanced material; mentor may go deeper depending on cohort interest.

Watch for student reactions at:

- The complexity bound for reverse-mode AD. Students with CS background will get this immediately; mathematicians may need it stated twice.
- The transition from grad/jacrev to vmap. The benefit is large and worth dwelling on.

Common Student Confusions

Confusions you should expect to see in office hours and recitation. The right-hand column gives suggested handling.

Confusion	How to Handle
Why are characteristic function methods faster than Black–Scholes for European options? Black–Scholes is closed form.	For European options under Black–Scholes, they are not faster — Black–Scholes is closed form and optimal. The point is that under Heston or any model where you do not have closed-form prices, characteristic function methods give you near-closed-form speed (FFT in log-K, COS as a

	series expansion).
What is the difference between forward-mode and reverse-mode AD?	Forward-mode: $O(\text{input dim})$ cost per output. Reverse-mode: $O(\text{output dim})$ cost per input. For Greeks (many outputs, few inputs), reverse-mode wins. Encourage students to derive the complexity bounds themselves.
My JAX implementation is slower than NumPy. Did I do something wrong?	Almost certainly: you are not using jit, or you are recompiling on every call. JAX traces and compiles; the first call is slow, subsequent calls are fast. Use <code>jax.jit</code> and ensure the function signature is static.
I cannot get my COS pricer to match Monte Carlo within $1e-4$. Help?	Most likely the truncation interval is wrong. Compute it from the cumulants of the characteristic function (the lecture covered this) rather than guessing. Second most likely: not enough COS terms; try $N = 2^{10}$ or higher.

Recitation Lesson Plan

Goal: Every student profiles a naive pricer, applies AD and vectorization, and measures at least one order of magnitude speedup. This is the dress rehearsal for the milestone.

Materials needed:

- A naive Python implementation of the COS pricer (no JAX, no vmap, just a loop), provided to students.
- A standardized portfolio of fifty Heston-priced options for the benchmark.
- A reference implementation showing the target performance (tutor only).

Suggested timing for the ninety-minute session:

Minutes	Activity
0–10	Recap. Show the latency budget: 5ms per surface revaluation, 200 revaluations per second, 300+ options per surface — that's a hard problem.
10–40	Profile the naive implementation. Each student identifies the bottleneck (usually the per-option price call inside a Python loop).
40–70	Convert to JAX with vmap and jit. Re-benchmark. Most students will see 10x to 100x speedup.
70–85	Add AAD: replace finite-difference Greeks with <code>jax.grad</code> . Re-benchmark. Most students will see another 5x to 10x.
85–90	Discuss what is left to optimize before the milestone.

Recitation-leader notes:

- Have the reference implementation timings ready; students need to know what "good" looks like.
- C++ via Enzyme is for the strong students. If one or two students are coasting, suggest the Enzyme exercise; do not require it.

Problem Set Solutions and Rubric

Solution sketches with point allocations. Tutors are expected to develop their own complete solutions before grading; the sketches below are calibration aids, not solutions manuals.

#	Pts	Key result expected	Common errors and partial credit
1	10	Carr–Madan FFT pricer for Heston European calls; convergence rate reported.	Wrong dampening parameter (α): -3. Failing to verify against Monte Carlo: -3.
2	10	COS pricer for the same problem; accuracy and runtime comparison with FFT.	Wrong truncation interval: -3. Failing to compare runtime: -2.
3	10	Forward-mode AD derived by hand for Black–Scholes call; verification against analytical Greeks.	Skipping the dual-number arithmetic: -4. Failing to verify: -3.
4	12	JAX-based full Greek vector for fifty options; runtime comparison with finite-difference.	Not using jit (slow): -3. Computing only first-order Greeks: -3.
5	8	JAX vmap vectorization; reported speedup in nanoseconds per evaluation.	Not reporting in the requested units: -2. Speedup of less than 10x: investigate, not a deduction per se.
6	10 (extra credit)	C++ via Enzyme: AAD performance comparison against JAX.	Optional. Award up to 10 points of extra credit; do not deduct from base grade.

Milestone Week Note

MILESTONE WEEK. Milestone 2 is due at the end of this week. See Part III, Section 2 for the full evaluation rubric. The latency target is non-negotiable: a library that revalues the surface in 5.5ms instead of 5ms does not pass the milestone. Be prepared for student arguments on this point; refer them to the rubric.

Week 8. Options Market Making Theory and Dynamic Hedging

THE SPINE OF THIS WEEK

The conceptual centerpiece of the program. Students must understand that options market making is GLFT with a vector-valued inventory, and they must understand how the resulting quoting strategy interacts with an explicit hedging policy. This is the most cognitively demanding week.

Lecture Emphasis

Below are notes for each of the two lectures. You do not deliver these; you attend them. Your job is to track where the cohort got lost so you can address it in recitation.

Lecture 1 — Multi-Dimensional Inventory and the Multi-Greek HJB

El Aoud and Abergel's paper, presented carefully. The lecture builds on GLFT by replacing the scalar inventory with a vector of Greeks and the scalar inventory penalty with a quadratic form. The mentor will derive the multi-Greek HJB and verify that it reduces to GLFT in the one-dimensional case. Students who have understood weeks 2–5 will follow; those who have not will struggle.

Watch for student reactions at:

- The introduction of the inventory vector. Students often need a moment to realize that they have been thinking of inventory as a scalar all program; it is.
- The penalty matrix. Some students will want to set it to the identity; the mentor will discourage this. Defer to office hours for the discussion of how to choose it.

Lecture 2 — Dynamic Hedging and Integration with Quoting

The practitioner lecture, drawing heavily on Taleb. The gamma-theta-variance identity is derived and forms the conceptual core. The integration of hedging with quoting is presented as a design choice — external versus integrated — with explicit pros and cons. The Coincall perpetual futures contract is discussed as the delta-hedging instrument.

Watch for student reactions at:

- The gamma-theta-variance identity. This is the conceptual key to options trading; students must understand it.
- The funding rate on perpetuals. Students from a traditional finance background may not have encountered this.

Common Student Confusions

Confusions you should expect to see in office hours and recitation. The right-hand column gives suggested handling.

Confusion	How to Handle
Why is the inventory a vector and not just a list of scalar positions?	Because the relevant risks are aggregated Greeks, not individual contract positions. A market maker who is long

	one call and short one call at different strikes has zero contract position but non-zero gamma and vega. The vector formulation captures this.
How do I choose the penalty matrix? Why not just the identity?	The identity penalizes a unit of delta exposure equally to a unit of gamma exposure. But the empirical risks of unit delta and unit gamma are very different. Calibrate the penalty matrix to the empirical covariance of the Greek processes, or to the desired risk-equivalence (e.g., one unit of delta risk equals one unit of vega risk at some volatility scale). The choice is part of the model design.
The gamma-theta-variance identity seems too simple. Is that really all there is to it?	Yes and no. The identity is exact under Black-Scholes; in reality, additional terms enter for stochastic volatility, jumps, and discrete hedging. But the identity is the right starting point and explains 80% of the daily P&L of a hedged book.
Should I delta-hedge with perpetual futures or with the underlying?	Perpetual futures, for two reasons: lower transaction costs (no settlement), and Coincall's options are quoted against the perpetual price index. The funding rate is a small but real cost that must be accounted for; it shows up in the milestone 3 backtest.

Recitation Lesson Plan

Goal: Students complete the multi-Greek HJB derivation and run a hedged-book simulation under two hedging policies (external threshold vs continuous), comparing the resulting P&L distributions.

Materials needed:

- Whiteboard for the derivation.
- A simulation scaffold for a hedged book under specified Heston dynamics.

Suggested timing for the ninety-minute session:

Minutes	Activity
0–35	Group derivation of the multi-Greek HJB. The reduction to GLFT in the one-dimensional case is the verification step at the end.
35–70	Pair programming. Each pair runs the hedged-book simulation under two policies. P&L distributions are aggregated on the projector.
70–85	Discussion. Which policy wins in which regime? Why? Tie to the design choice for milestone 3.
85–90	Wrap-up. Preview of milestone 3 and the role of the perpetual futures contract.

Recitation-leader notes:

- This is the second-most-important whiteboard derivation in the program (after week 2). Move slowly enough that every student is following.

Problem Set Solutions and Rubric

Solution sketches with point allocations. Tutors are expected to develop their own complete solutions before grading; the sketches below are calibration aids, not solutions manuals.

#	Pts	Key result expected	Common errors and partial credit
1	10	Inventory dynamics for a portfolio of options; jump structure on fills.	Missing the multi-contract structure: -3. Wrong Greek aggregation: -3.
2	15	El Aoud–Abergel HJB derivation; separation ansatz; reduction to PDE on inventory vector; reduction to GLFT in 1D limit.	Skipping the verification that it reduces to GLFT: -4. Algebra errors in the multi-dimensional case: -3 each.
3	10	Penalty matrix calibrated to Coincall data; entries justified from empirical Greek covariances.	Using the identity matrix: -4. Failing to justify entries from data: -4.
4	10	Gamma–theta–variance identity derived for a continuously hedged short call.	Wrong sign on theta: -2. Missing the half-factor in front of gamma: -2. Failing to identify the realized-vs-implied variance term: -3.
5	10	Threshold delta-hedging policy implemented on one week of Coincall data; hedging cost and risk reduction reported; integration with GLFT engine.	Missing the integration with GLFT: -4. Reporting only cost without risk reduction: -3.
6	8	Three modeling assumptions in El Aoud–Abergel critically discussed for crypto.	Listing without proposing modifications: -3. Modifications that are not actually implementable: -2.

Week 9. Backtesting and Risk Analysis

THE SPINE OF THIS WEEK

Students assemble everything they have built into a working backtest on three months of Coincall data. The week is more about engineering discipline than new theory; the lecture material is necessary but not the centerpiece.

Lecture Emphasis

Below are notes for each of the two lectures. You do not deliver these; you attend them. Your job is to track where the cohort got lost so you can address it in recitation.

Lecture 1 — Backtesting Architecture for Options

The lecture covers the architecture of an event-driven options backtest and emphasizes the pitfalls. The mentor will list common bugs: look-ahead bias, unrealistic fill assumptions, expiry handling, contract roll. Students should leave knowing what to test for, not just what to build.

Watch for student reactions at:

- The fill simulator design. Students will be tempted to assume 100% fill at their quoted prices; the mentor will explain why this is wrong.
- The contract expiry handling. Options are not perpetual; there are discrete events when a series expires.

Lecture 2 — Risk Metrics and P&L Attribution

Sharpe, drawdown, Calmar, and the like are reviewed quickly. The substantive content of the lecture is P&L attribution: decomposing daily P&L into spread, inventory drift, gamma, vega, and hedging streams. This decomposition is the heart of the milestone deliverable.

Watch for student reactions at:

- The attribution sum-to-total check. Students sometimes have attribution streams that do not sum to total P&L; this is always a bug.

Common Student Confusions

Confusions you should expect to see in office hours and recitation. The right-hand column gives suggested handling.

Confusion	How to Handle
My backtest P&L is suspiciously high. Bug or genuine alpha?	Bug. Always bug. Common culprits: look-ahead in the volatility surface (you used today's surface to quote yesterday's snapshot), 100% fill assumption, missing transaction costs, missing funding rate on perpetual hedges. Walk through the data flow one event at a time.
How do I calibrate the fill simulator? It	Use a holdout period. Calibrate the fill model on one

seems circular.	month, validate on a non-overlapping month. The key empirical fact: fill probability is approximately exponential in quote distance from mid, with a parameter that depends on contract liquidity.
The P&L attribution sums to a different number than the total P&L. Where's the bug?	Almost certainly: you are computing the streams using different state snapshots. The spread stream uses end-of-trade state; the inventory drift stream should use start-of-period state. Get the timing right and they will sum.
My Sharpe ratio is 6.0 over three months. Is that real?	No. Sharpe over a three-month window has large estimation error. The deflated Sharpe ratio (Bailey and López de Prado 2014) corrects for this; require the student to report a confidence interval from block bootstrap.

Recitation Lesson Plan

Goal: Each student presents a one-week regime from their backtest, walks through the P&L attribution, and answers questions from the cohort. This is the dress rehearsal for the milestone.

Materials needed:

- Time slot for each student to present (8 students $\times$ 10 minutes = 80 minutes, leaving room for transitions).
- A projector for each presentation.

Suggested timing for the ninety-minute session:

Minutes	Activity
0–5	Set the agenda. Each student gets 8 minutes (5 to present, 3 for Q&A).
5–85	Eight presentations, back to back.
85–90	Wrap-up: what common issues did we see? Refer everyone to the milestone rubric.

Recitation-leader notes:

- If the cohort is smaller, give more time per student. If the cohort is larger, do this over two sessions.
- The presentations should expose engineering bugs that students will then fix before the milestone. This is the entire purpose of the recitation.

Problem Set Solutions and Rubric

Solution sketches with point allocations. Tutors are expected to develop their own complete solutions before grading; the sketches below are calibration aids, not solutions manuals.

#	Pts	Key result expected	Common errors and partial credit
1	12	Event-driven backtest harness with deterministic replay verified by bitwise-identical re-run.	Non-deterministic results: -5. Code that runs but with floating-point drift: -2 (warn the student).
2	12	Fill simulator calibrated on training period, validated on holdout.	No holdout validation: -5. Overfitting (training fits great, holdout poor) without commentary: -3.
3	12	Full one-month backtest; performance metrics reported with appropriate caveats.	Missing block-bootstrap confidence intervals: -3. Reporting Sharpe without context (e.g., compared to a baseline): -2.
4	10	P&L attribution decomposition; stacked-area plot of cumulative attribution.	Attribution that does not sum to total P&L: -5 (this is always a bug). Aesthetics: not graded, but flag bad plots in feedback.
5	10	Worst-drawdown day forensic analysis.	Forensic that does not identify specific positions: -4. Forensic that lacks counterfactual analysis: -3.

Milestone Week Note

MILESTONE WEEK. Milestone 3 is due at the end of this week. See Part III, Section 3 for the full evaluation rubric. This is the most consequential milestone of the program and the basis for the final report. Code review sessions should run six to eight hours total (in addition to your usual office hours).

Week 10. Extensions, Final Report, and Presentations

THE SPINE OF THIS WEEK

Students choose one extension topic, implement it in a week, and present the final report. The tutor's role shifts from instruction to mentorship: helping each student scope their extension realistically and giving feedback on the draft report.

Lecture Emphasis

Below are notes for each of the two lectures. You do not deliver these; you attend them. Your job is to track where the cohort got lost so you can address it in recitation.

Lecture 1 — Extension Topics

The mentor surveys the menu of extension topics, sketches what a credible week of work looks like for each, and answers questions. Students choose by end of day. Tutor: pay attention to who picks what; you will be the first line of consultation as they implement.

Watch for student reactions at:

- Students choosing topics that are too ambitious. The mentor will guide, but the tutor should reinforce: a clean partial result beats an incomplete grand result.

Lecture 2 — Research Communication

The mentor walks through the structure of a quantitative research paper using examples. The lecture is short on theory and long on examples; encourage students to take notes on style.

Watch for student reactions at:

- Students from a pure-math background often resist the "acknowledge limitations" advice. Re-emphasize in feedback: this is industry research writing, not a theorem-proof paper.

Common Student Confusions

Confusions you should expect to see in office hours and recitation. The right-hand column gives suggested handling.

Confusion	How to Handle
My extension is not working. Should I switch?	Probably not. The week is too short to switch and start over. Identify the smallest version of your extension that does work, write that up as a partial result, and acknowledge the limitations honestly in the report.
How long should my final report be?	Maximum thirty pages excluding appendices, but quality beats length. A tight twenty-page report is preferable to a sprawling thirty-page one. Encourage students to write a short report and use the appendix for supporting material.

Can I use generated text or LLM assistance in the report?	Refer to the academic integrity policy in Part IV. Short version: use LLMs as a writing tool (grammar, phrasing), not as a content generator. All technical content must be the student's own work.
How long should my presentation be?	Twenty minutes including questions. Practice it once with a tutor before the final session.

Recitation Lesson Plan

Goal: Each student delivers a fifteen-minute dry-run presentation. The cohort provides feedback. The tutor focuses on questions an industry audience would ask.

Materials needed:

- Projector.
- Stopwatch.
- Feedback rubric (provided in Part III, Section 4) for the cohort to fill in for each presenter.

Suggested timing for the ninety-minute session:

Minutes	Activity
0–5	Set expectations: this is a dry run, not the final presentation. Tutor will be tough.
5–85	Eight presentations, 10 minutes each (presentation + Q&A).
85–90	Aggregate feedback: common issues, common strengths.

Recitation-leader notes:

- Be tough but constructive. The point of the dry run is to expose problems while there is still time to fix them.
- Submit the feedback rubrics to the mentor; they inform the final presentation grade.

Problem Set Solutions and Rubric

Solution sketches with point allocations. Tutors are expected to develop their own complete solutions before grading; the sketches below are calibration aids, not solutions manuals.

#	Pts	Key result expected	Common errors and partial credit
1	5	One-page extension proposal submitted by end of day one of week 10.	Late submission: -3. Proposal that does not engage with the realistic time budget: -2.
2	20	Implementation of the chosen extension on top of milestone infrastructure.	Extension that does not actually run: at most 8 points for design. Partial

			implementations that are honestly reported: scaled credit.
3	15	Quantitative comparison of extension against week-9 baseline using shared metrics.	Comparison missing baseline: -5. Cherry-picked metrics: -5.
4	25	Final technical report (max 30 pp.) following the lecture's structure.	See the report rubric in Part III, Section 4.
5	15	Final twenty-minute presentation.	See the presentation rubric in Part III, Section 4.

Part III. Milestone Evaluation Rubrics

Each milestone is graded out of 100 points. Final grade for the milestone is the average of two independent evaluations: yours as the tutor, and the mentor's. Significant discrepancies (more than 10 points) trigger a third reviewer.

3.1 Milestone 1 — Linear-Asset GLFT Engine

Due: end of week 5. Evaluation criteria sum to 100 points.

Criterion	Points	Description
Correctness of GLFT implementation	30	GLFT quote computation produces results that match the closed-form asymptotic formula within numerical tolerance ($1e-6$). Verified by running the engine on a synthetic problem with known analytical solution.
Intensity calibration quality	20	MLE-based calibration produces a non-trivial fit with χ^2 goodness-of-fit reported. The student must demonstrate awareness of the limitations of their fit, not just produce numbers.
Backtest reproducibility	15	Running the backtest twice with the same configuration produces bitwise-identical results. Tested by the grader; non-reproducibility is an automatic -15.
Backtest performance match	15	P&L within $\pm 50\%$ of the mentor's reference implementation on the same data window. The 50% tolerance is generous because of legitimate implementation choices; outside this band suggests a bug.
Latency	10	GLFT quote computation in under $100\mu s$ per update on a single CPU core. Measured by the grader using a standardized benchmark script.
Code quality	5	Code passes linting, includes type hints, and has unit tests covering the core quote computation. Subjective but not capricious.
Technical report	5	Report is at most 10 pp., follows the structure of the GLFT paper, and acknowledges limitations honestly.

Common reasons for partial credit:

- Code runs but uses finite-difference Greeks (which we have not yet introduced): -5 from correctness, since the implementation is sub-optimal but not wrong.
- Reproducibility fails due to a random seed not being set: -8, with explicit feedback on how to fix.
- Report exceeds 10 pages: -2 per additional page, up to -5.

3.2 Milestone 2 — Fast Pricing and Greeks Library

Due: end of week 7. Evaluation criteria sum to 100 points.

Criterion	Points	Description
Pricing accuracy	20	Maximum relative error of $1e-4$ against the Coincall reference prices across all contracts in the BTC surface. Measured by the grader.
Greek accuracy	15	Finite-difference verification of all six Greeks (delta, gamma, vega, theta, vanna, volga) against the AAD-computed Greeks, with relative error below $1e-4$.
Latency — full surface revaluation	25	Full Coincall BTC surface (300-500 contracts) revalues with full Greek vector in under 5ms on a single CPU core. Measured by the grader. THIS IS A HARD GATE: failing this criterion caps the milestone grade at 70.
Latency — per-option breakdown	10	Per-option latency in nanoseconds reported with distribution (mean, median, p95, p99). Bonus if students report tail latencies that are bounded.
API design	10	Library exposes a clean Python API. Importable as a package. Sensible function signatures. Documentation strings.
Unit tests	10	Tests cover all pricers, all Greeks, the calibrated surface adapter, and at least one edge case (e.g., extreme moneyness, zero time-to-expiry).
Technical note	10	Benchmarking results documented with latency distributions (histograms, not just means), and a clear explanation of optimization choices.

Common reasons for partial credit:

- Latency target missed by less than 1ms (i.e., 5–6ms instead of 5ms): -10 from the latency criterion but no hard cap. Beyond 6ms: hard cap applies.
- Library works in JAX but does not generalize beyond Black–Scholes: -8 from API design.
- Vanna and volga not implemented (only first-order Greeks): -8 from Greek accuracy.
- C++ Enzyme exercise completed: +5 extra credit (capped at 100 total).

3.3 Milestone 3 — Full Options Market Making Backtest

Due: end of week 9. Evaluation criteria sum to 100 points. This is the most consequential milestone and the basis for the final report; grade carefully.

Criterion	Points	Description
Backtest reproducibility	10	Identical replay produces bitwise-identical results. As in milestone 1, non-reproducibility is an automatic -10.
P&L attribution correctness	20	Attribution streams (spread, inventory, gamma, vega, hedging) sum to total P&L within 1bp tolerance. Failure here usually indicates a timing bug in the attribution code.
Risk metrics	15	All required metrics (Sharpe, max drawdown, Calmar) reported with block-bootstrap confidence intervals. The CI requirement is non-negotiable.
Greek exposure tracking	10	Time series of aggregated Greeks (delta, gamma, vega) plotted; exposures remain within configured position limits.
Engine quality	15	Multi-Greek market making engine correctly integrates the calibrated volatility surface (milestone 2 library) with the GLFT framework (milestone 1).
Hedging integration	10	Dynamic hedging policy correctly integrated; hedging cost accounted for (including funding rate on perpetuals).
Forensic analysis	10	Drawdown forensic identifies specific positions, the market environment, and at least one counterfactual.
Technical report	10	Report is at most 20 pp., includes the forensic analysis, and acknowledges failure modes honestly.

Common reasons for partial credit:

- Look-ahead in the volatility surface: -15 from engine quality. Very common bug; check carefully.
- 100% fill assumption: -10 from engine quality. Almost as common.
- Funding rate ignored: -5 from hedging integration.

- Sharpe ratio reported without CI: -8 from risk metrics.

3.4 Final Report and Presentation

Final report: 25 points. Final presentation: 15 points. Total of 40 points; combined with the three milestones and the problem sets to produce the final grade.

Report rubric:

Criterion	Pts	Description
Problem motivation	3	Clearly states what the extension addresses and why it matters. A reader from industry should understand the motivation in two paragraphs.
Methodology	6	Methodology is technically sound, builds on the milestone infrastructure, and is described at the right level of detail (enough to reproduce; not so much that it overwhelms).
Results	6	Empirical results are reported with appropriate uncertainty quantification. Plots are clear and labeled.
Discussion	4	Discussion engages honestly with the results. Successes are claimed without overstatement; failures are acknowledged and analyzed.
Limitations	3	An explicit limitations section. This is required, not optional.
Writing quality	3	Prose is clear and free of obvious errors. Citations are correct.

Presentation rubric:

Criterion	Pts	Description
Structure	3	Presentation follows a clear structure: motivation, methodology, results, discussion, conclusions. Time is allocated appropriately.
Technical depth	4	Technical content is correct and at the right level for the audience. The presenter can answer follow-up questions.
Visual aids	3	Slides are clear, uncluttered, and well-labeled. No more than one main idea per slide.
Delivery	3	Pacing is appropriate; the presenter is audible; eye contact is maintained.
Q&A	2	The presenter handles questions gracefully, including questions they cannot fully answer.

Part IV. Administrative Reference

4.1 Academic Integrity

The program follows the NC State Code of Student Conduct on academic integrity. Specific guidance for this program:

- Collaboration on problem sets is encouraged but submissions must be each student's own work. Acceptable: discussing the approach. Unacceptable: copying code or derivations.
- Citation is required for any external source used in a problem set or report. "External" includes papers, textbooks, blog posts, and LLM-generated content.
- Use of LLM assistance is allowed for writing support (grammar, phrasing) but not for technical content generation. Students should disclose in the report any non-trivial LLM use.
- The mentor handles all academic integrity cases. If you suspect a violation, document it (specific evidence, dates) and escalate to the mentor within twenty-four hours. Do not confront the student yourself.

4.2 Late Work

Late submission policy for problem sets:

- Submissions are due Sunday at 11:59pm Eastern time.
- Late submissions are accepted until Tuesday 11:59pm Eastern with a 20% deduction.
- Submissions later than Tuesday are not accepted; the student receives zero for that problem set.
- Exceptions for documented medical or family emergencies are handled by the mentor, not the tutor. Direct the student to the mentor.

Late submission policy for milestone deliverables:

- Milestones are due Friday at 11:59pm Eastern time of the milestone week.
- Late milestones are not accepted. The student receives zero for that milestone.
- Exceptions are again handled by the mentor only.

4.3 Accommodations

Students with documented accommodations (additional time, alternative formats, etc.) should have their accommodations on file with the NC State Disability Resource Office. Tutors implement accommodations as documented; do not ask the student to justify them. If you are uncertain how to implement a specific accommodation, ask the mentor.

4.4 Crisis Escalation

Situations to escalate to the mentor immediately:

- Suspected academic integrity violation.
- Student in apparent acute distress (mental health, family emergency, etc.).

- Data security incident (e.g., student inadvertently exposed Coincall data on a public repository).
- Hostile interaction between students or between you and a student.

Use the emergency phone line for these; do not wait on Slack. The mentor will direct next steps. Do not attempt to resolve these yourself.

4.5 Office Setup and Logistics

Each tutor is provided with:

- Access to the shared Slack workspace.
- Access to the NC State LMS for the program (for problem set submission and grade entry).
- Access to the Coincall data S3 endpoint (under the same NDA as students, plus additional credentials for the reference implementations).
- A laptop with the standard program environment pre-installed: Python 3.11, JAX, NumPy, SciPy, Pandas, PyArrow, QuantLib, and the program's own internal libraries.
- A desk in the shared tutor office in SAS Hall (key issued at orientation).

Setup tasks for the first week:

1. Complete the data use agreement with the program coordinator.
2. Verify your Slack and LMS access.
3. Run the reference Milestone 1 implementation end-to-end to confirm your environment is set up correctly.
4. Attend the tutor calibration session.

4.6 Compensation and Hours

Tutors are compensated as standard graduate teaching assistants under the NC State pay scale for the Department of Mathematics. Hours and compensation are tracked through the standard university time-reporting system.

4.7 Contact List

Maintained separately as a one-page reference distributed at orientation. Includes: program mentor, program coordinator, departmental administrator, IT support, and emergency contacts. Keep accessible during the program.